Technology Stack

Date	7 July 2026
Team ID	SWTID -2026-4213
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*Streamlit (Python)*	*Provides a simple, interactive, and responsive user interface for entering event details, generating conversation starters, viewing history, and submitting feedback with minimal frontend development effort.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	FastAPI(*Python)*	*Handles REST APIs, processes user requests efficiently, connects AI models, and provides fast, scalable backend services with automatic API documentation.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*JSON* Files *(history.json, feedback.json)*	*Lightweight local storage for saving conversation history and user feed-back without requiring a database server, making it suitable for an academic project.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Localhost(Uvicorn +Streamlit)*	*The application is deployed locally for development, testing, and demonstration purposes with easy setup and execution.*
5	**Version Control & CI/CD**	*Git & Github*	*Manages source code, tracks version history, supports collaboration, and serves as the project submission repository.*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Wikipedia API, Hugging Face Transformers (DistilBERT & GPT-2),PyTorch*	*Wikipedia API verifies facts, DistilBERT analyzes event themes, GPT-2 generates personalized conversation starters, and PyTorch provides AI model execution.*